Ayushi Rathod

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Technical Skills

- **Programming & Data Analysis:** Python (pandas), R, SQL, Excel, VBA, Power Query M, Jupyter Notebooks.
- **Analytics & Reporting:** Data Processing, KPI Reporting, Trend Analysis, Data Integrity, Stakeholder Insights.
- **Visualization Tools:** Power BI (DAX, data modelling), Tableau, Excel Dashboards, Interactive Reporting.
- **Statistics:** Regression, Hypothesis Testing, Forecasting, Data Interpretation, Measurement Analysis.
- **Data Management:** ETL Pipelines, Data Cleaning, SQL (MySQL, PostgreSQL), Data Validation, ERD Design.
- **Automation & Tools:** Power Automate, Power Apps, Git/GitHub, Azure, AWS, BigQuery.

Education

Northeastern University - Toronto, Canada *Sep 2024 – Mar 2026*
Master of Professional Studies in Analytics

- **Coursework:** Machine Learning, AI, Statistical Modeling, Data Visualization, Business Intelligence.

The Maharaja Sayajirao University of Baroda *Jul 2021 – May 2023*
Master of Computer Applications (MCA)

The Maharaja Sayajirao University of Baroda *Jun 2018 – May 2021*
Bachelor of Science in Mathematics

Professional Experience

Confidosoft Solutions - Vadodara, India *Jan 2023 – Jul 2024*
Data Analyst & Business Intelligence Associate

- Managed end-to-end data-processing workflows ensuring data accuracy, integrity, and consistency across reports.
- Extracted, cleaned, and transformed large datasets using SQL, Python, and Power Query for analysis readiness.
- Developed Power BI dashboards with DAX to visualize KPIs, trends, and performance insights for stakeholders.
- Automated reporting workflows using VBA, Power Query, and Python reducing manual effort by 40%.
- Applied statistical analysis to identify patterns, trends, and drivers impacting business performance.
- Collaborated with cross-functional teams to gather requirements and align insights with business objectives.
- Built reusable SQL queries and data models to standardize reporting and improve data reliability.
- Screened and prioritized relevant datasets to support data-driven decision-making and reporting efficiency.
- Documented data definitions, transformations, and workflows to support governance and auditability.
- Improved reporting turnaround time by 25% through process optimization and automation.

Projects

- **Behavioral Health Risk Scoring System (Python, BigQuery & ML):**
  - Built machine learning models (XGBoost, Random Forest) on 400K+ records achieving 87% accuracy for risk classification.
  - Engineered a composite 0–100 risk score using feature scaling and normalization for effective risk stratification.
  - Applied SHAP for model interpretability to identify key behavioral drivers influencing high-risk outcomes.
  - Developed scalable data pipelines in BigQuery for automated data preprocessing and efficient data handling.
  - Enabled data-driven interventions improving decision-making efficiency and risk identification accuracy by 25%.
- **Customer Churn Prediction Model (Python, SQL & Power BI):**
  - Developed classification models (Logistic Regression, Random Forest) achieving 85% accuracy and strong ROC-AUC performance.
  - Engineered behavioral and transactional features to enhance predictive signal and model performance.
  - Handled class imbalance using SMOTE and class weighting to improve recall and model robustness.
  - Evaluated models using precision, recall, F1-score, and ROC-AUC to ensure optimal model selection.
  - Delivered actionable insights via Power BI dashboards and SHAP visualizations improving retention strategies by 20%.
- **Operational KPI Dashboard (Power BI, SQL & Python):**
  - Designed end-to-end Power BI dashboard integrating SQL and Python, reducing manual reporting effort by 40%.
  - Built automated data pipelines enabling real-time refresh and improving decision-making speed by 30%.
  - Performed data cleaning and transformation ensuring cross-system data accuracy and consistency.
  - Conducted trend analysis to identify operational inefficiencies and support performance improvements.
  - Improved dashboard usability and stakeholder adoption, increasing insight-driven decisions by 25%.